



GRIDGUARD AI

AI-Based LT Line Fault Detection Automatic Emergency Shutdown System

Report ID	GG-20260716-195138
Generated On	16-07-2026 19:51:38
AI Prediction	Normal
Confidence	100.00%
Grid Health	100%
Emergency Shutdown	NO

Executive Summary

GridGuard AI continuously monitors the Low Tension (LT) electrical distribution network using Artificial Intelligence. During this monitoring cycle, the AI detected the fault **Normal** with a confidence level of **100.00%**. Based on the severity of the detected condition, the Automatic Emergency Shutdown mechanism was **NO** to protect transformers, feeders and consumers. Current Grid Health: **100%**

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Executive Dashboard

Grid Health	AI Confidence
100 %	100.00%
Detected Fault	Emergency Shutdown
Normal	NO
Overall Risk	
LOW	

Executive Summary Dashboard

GRID HEALTH	AI CONFIDENCE	FAULT
100 %	100.00 %	Normal
SHUTDOWN	RISK LEVEL	STATUS
NO	LOW	ACTIVE

Live Sensor Values

Voltage (V)	230.0
Current (A)	5.0
Frequency (Hz)	50.0
Temperature (°C)	30.0

Executive Decision

Grid is operating normally. AI recommends continuous monitoring. Preventive maintenance schedule remains unchanged.

Parameter	Value
Voltage (V)	230.0
Current (A)	5.0
Frequency (Hz)	50.0
Temperature (°C)	30.0
Detected Fault	Normal
AI Confidence	100.00%
Grid Health	100%

AI Prediction

Detected Fault : Normal

Prediction Confidence : 100.00%

Grid Health : 100%

Emergency Shutdown : NO

AI Recommendation

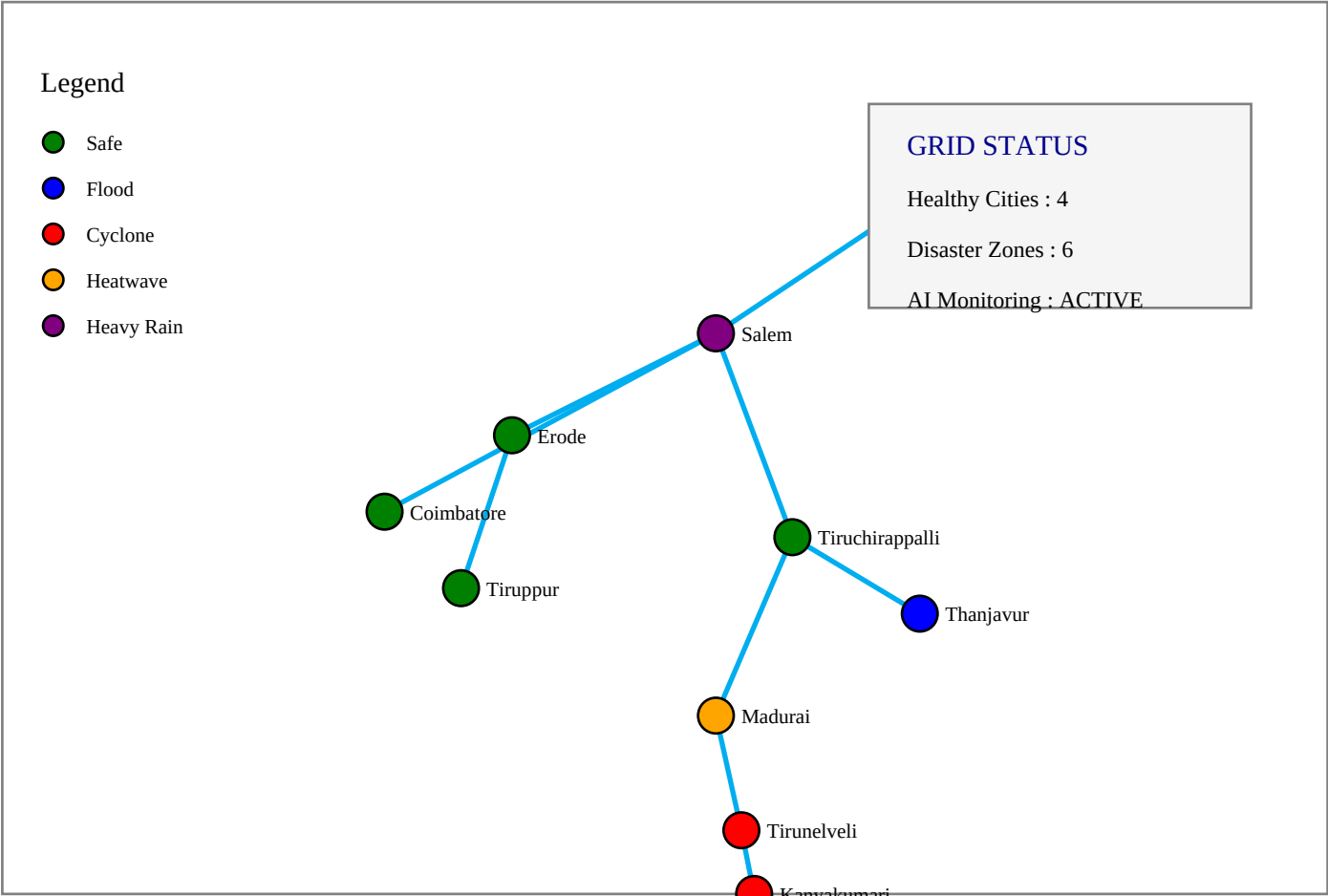
- ✓ Grid operating normally.
- ✓ Continue monitoring.
- ✓ Preventive maintenance every 30 days.

Grid Status

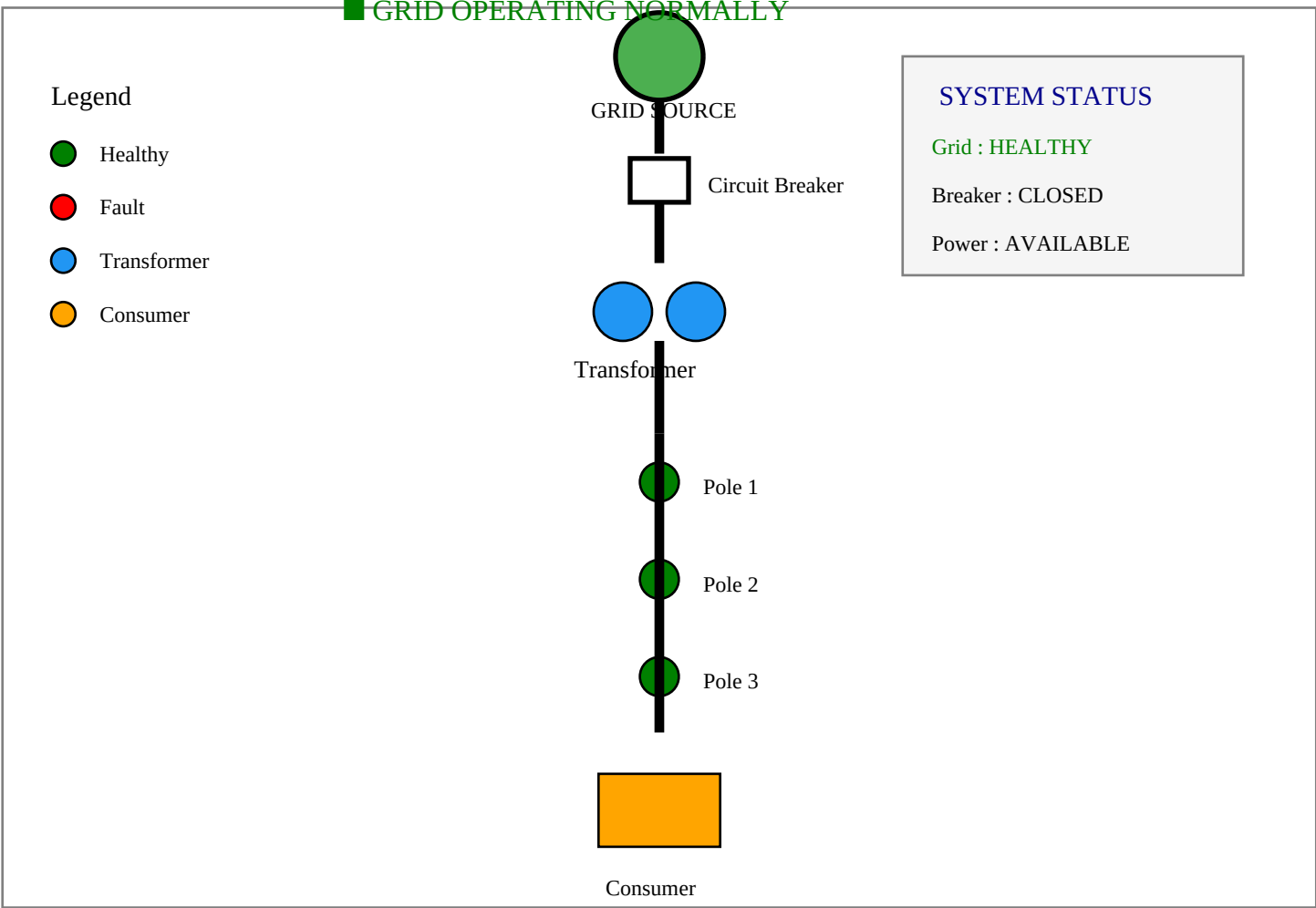
■ **GRID OPERATING NORMALLY**

Tamil Nadu Power Grid GIS

TAMIL NADU SMART GRID GIS

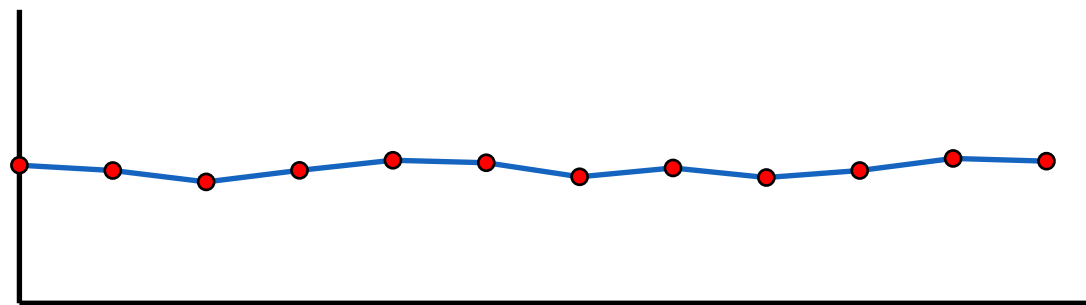


SMART GRID SINGLE LINE DIAGRAM

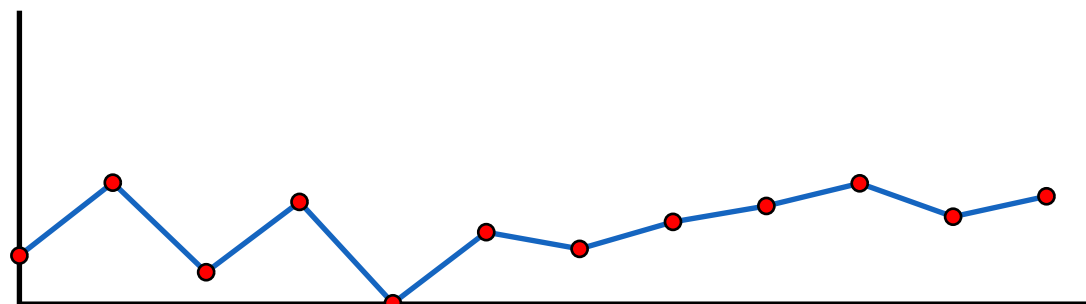


Sensor Trend Analysis

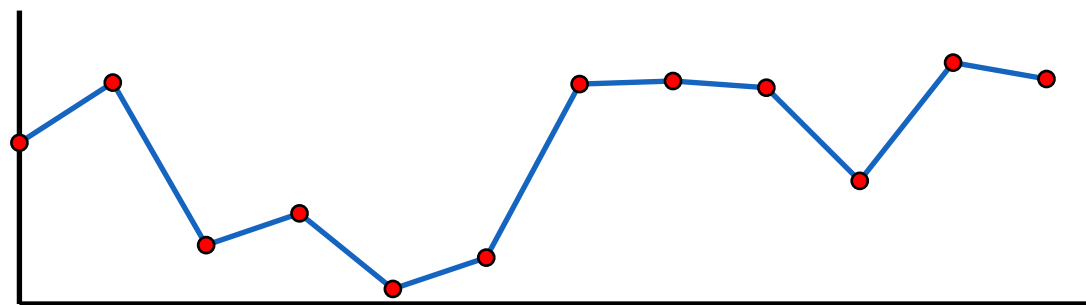
Voltage



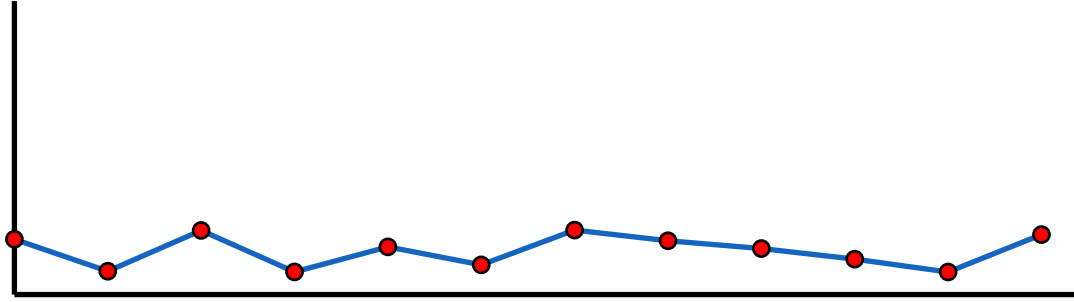
Current



Frequency



Temperature



Fault Event Timeline

Time	Event
00:00	Sensor Reading
00:01	AI Fault Prediction
00:02	Risk Analysis
00:03	Emergency Shutdown
00:04	SCADA Updated
00:05	PDF Report Generated

Application QR Code



Report Verification

Report ID : GG-20260716-195138

Generated By : GridGuard AI

Software Version : Version 3.0

Generated On : 16-07-2026 19:51:38

✓ **VERIFIED REPORT**

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